

Introduction

Project motivation: London’s growing population requires better public transport

- Improve efficiency of the transport system
- Reduce car dependency and carbon emissions

Approach:

- Graph theory analysis of London’s train network
- Assess network robustness and stability
- Suggest potential improvements

Methodology

I used three main techniques for this project:

- Data analysis
- K-edge connectivity and augmentation
- Random and targeted station removal

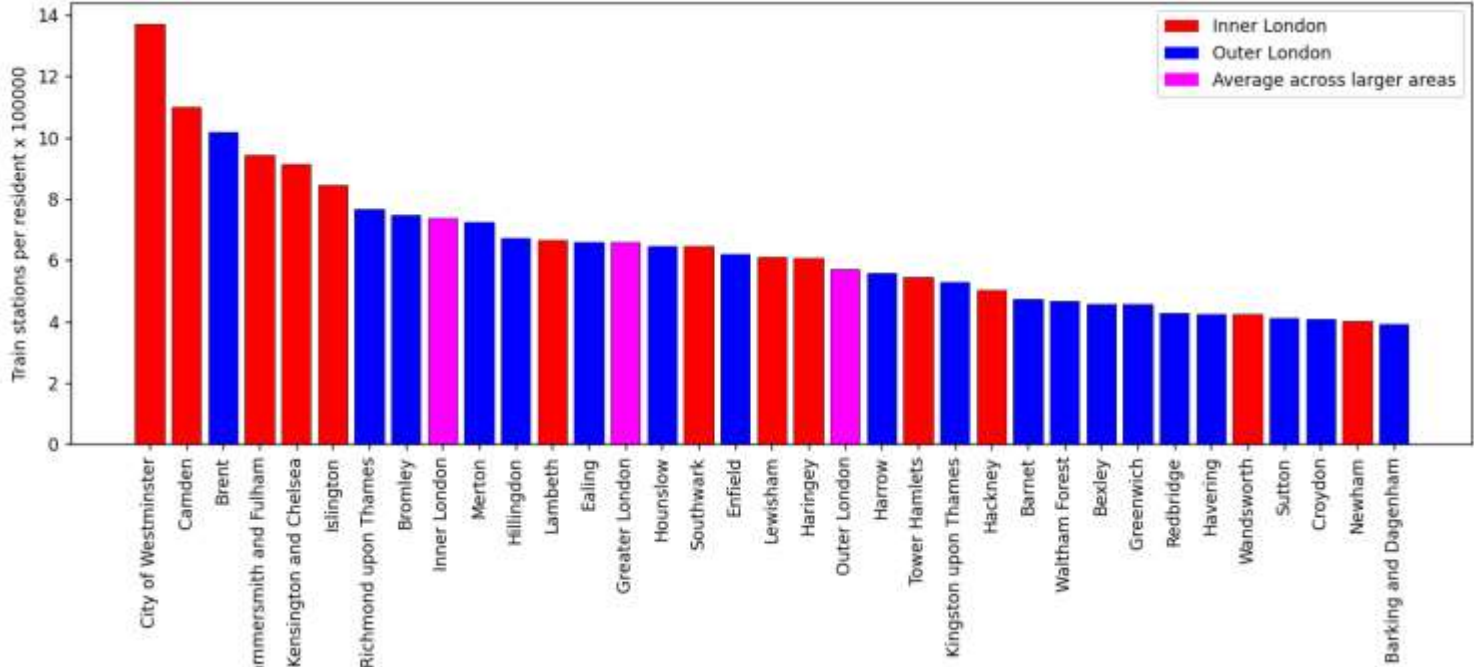
Part 1 – Data analysis

Examines population and transport distribution in Greater London

Population vs station density:

- Identifies areas with high population but few stations

Purpose – highlight where public transport improvements are needed



A bar chart showing train station density per resident per borough

Part 2 – K-edge connectivity and augmentation

Network modelling: London train network represented as a graph

- Stations = nodes, Tracks = edges

K-edge connectivity: Measures network resilience to failures

K-edge augmentation: Adding edges to achieve k-edge connectivity

Applied to various network scenarios

Goal: Find minimum total edge length to achieve k-edge connectivity

Used to compare effectiveness of different network designs

Part 3 – Random and Targeted station removal

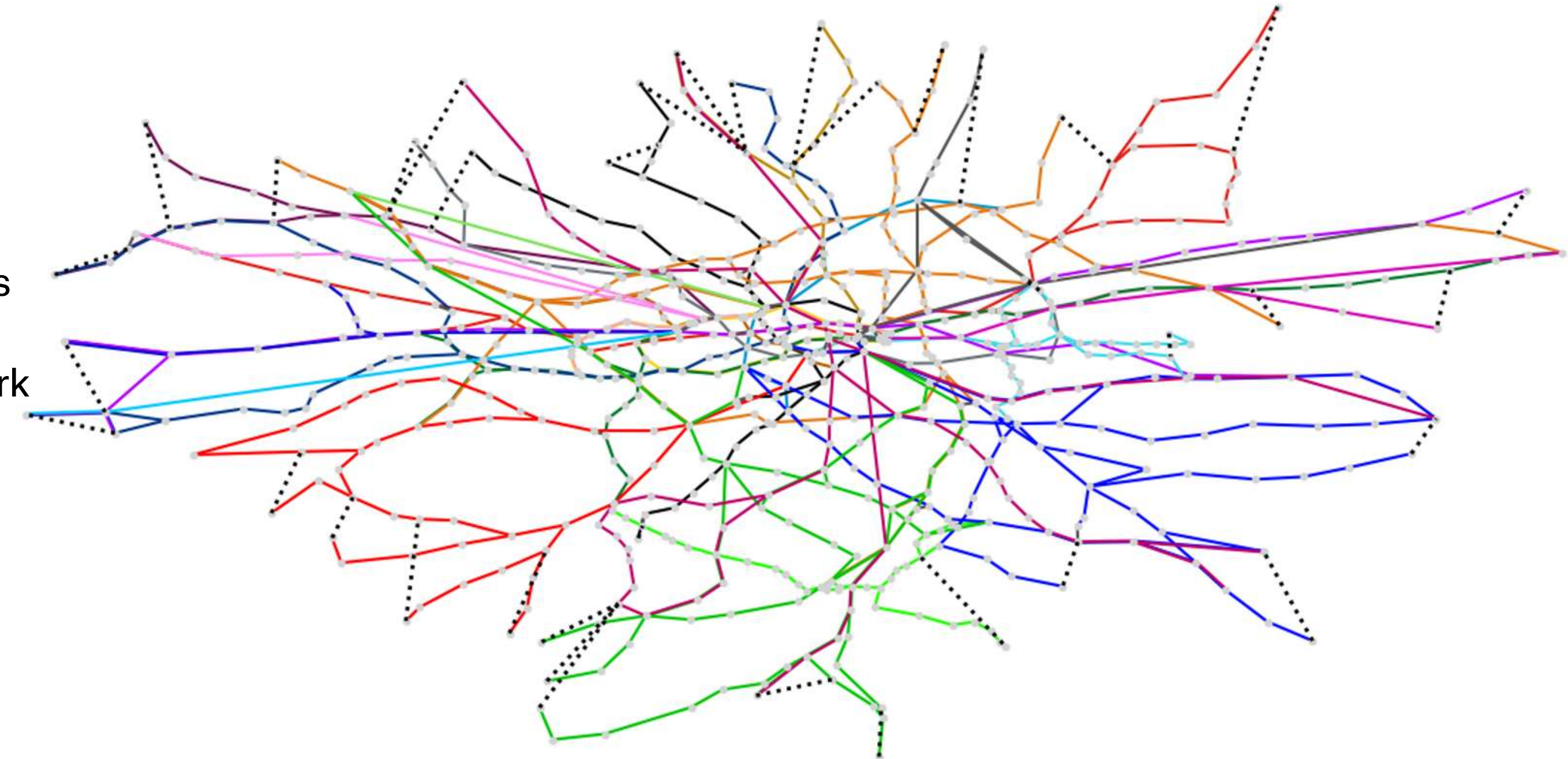
Examines impact on network connectivity

Random removal: Simulates real-world disruptions

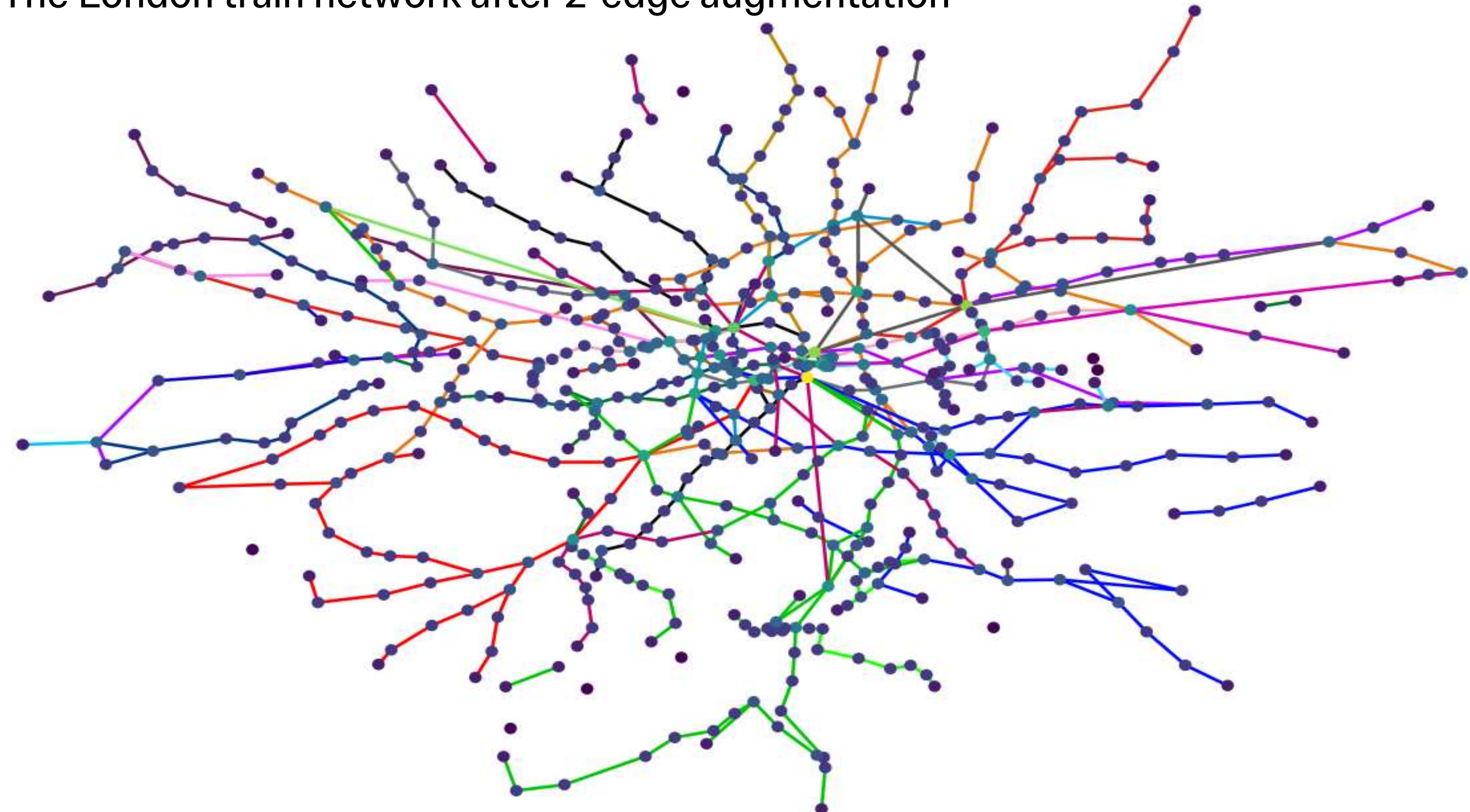
- Signalling faults, track maintenance, other delays

Targeted removal: Tests network under worst-case scenarios

Identifies vulnerabilities in key stations



The London train network after 2-edge augmentation



The London train network after 10% station removal

Results

Scenario	K = 2	K = 3	K = 4
Zones 1-6	37,114km	298,365km	564,748km
Zones 1-6 with superloop	38,104km	297,359km	559,732km
Zones 1-6 with proposed lines	37,107km	292,355km	556,735km
Zones 1-6 with proposed + superloop	36,96km	287,345km	548,711km
Zones 1-9	37,111km	312,435km	581,861km
Zones 1-9 with superloop	38,104km	309,425km	582,846km
Zones 1-9 with proposed lines	38,113km	309,434km	577,860km
Zones 1-9 with proposed + superloop	36,103km	304,418km	577,849km

- Shown left:
- Table summarising K-edge augmentation results
 - Graphs for random and targeted station removal:
 - Average number of connected subgraphs after random removal
 - Number of subgraphs after removing top x% of the most connected stations

Conclusions

- Network is robust to random failures but targeted attacks pose serious risks
- Buses very important to connectivity
- Even a small number of added edges contributes significantly to connectivity

Future work could include:

- Analysing bus network, cycle lanes and walking routes
- Analysing travel times

Could be used to analyse multimodal connectivity and suggest further improvements

Acknowledgements

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